SE(3)-WAM: Role-Aware 4D World–Action Modeling for Dynamic Manipulation

Anonymous CVPR submission

Abstract

Manipulating moving objects requires actions that fit the scene when they execute, not when the image was captured, yet every millisecond spent inferring the scene's motion lets the target move further. World–action models forecast future images or latents at uniform fidelity, spending time on appearance while leaving the target's metric pose implicit; target-centric predictors are fast but ignore the surroundings that constrain or drive the motion. We observe that the parts of a dynamic scene need different precision: the target's pose decides contact to within millimeters, the bodies around it matter only through the clearance they leave and the motion they impart to the target, and the remaining background does not matter. We present SE(3)-WAM, a world–action model built on this role-aware 4D structure. It encodes the target's motion as a metric SE(3) pose history, so one RGB-D frame suffices as input. It learns to forecast the target's future SE(3) trajectory precisely and the whole scene coarsely as sparse 3D flow, forecasts used only in training. Together, these choices let SE(3)-WAM act in [XX] ms on one RTX 5090, [N]× faster than video world–action baselines. We also introduce Dynamic-14, an open-source benchmark of 14 simulated dynamic manipulation tasks in which the scene keeps moving throughout each policy's inference. SE(3)-WAM achieves [XX.X]% success on Dynamic-14, [XX.X] points above the strongest baseline, and [XX]% across six real-robot tasks with diverse target motion.

1. Introduction

Learning-based manipulation has advanced rapidly, from diffusion policies to vision–language–action (VLA) and world–action models (WAMs) [6, 7, 12, 20, 40, 46, 60], yet most of this progress assumes that objects stay still unless the robot moves them. In factories, kitchens, and human–robot handovers, objects keep moving while the robot perceives, computes, and acts. In such scenes, an action must fit the scene at the moment it executes, not the moment its observation was captured, so a policy has to know how the scene moves. That knowledge must itself be inferred, and inference takes time during which the scene keeps moving: the longer a policy computes to anticipate the motion, the further ahead it must anticipate.

Recent dynamic policies add a memory of past motion, typically image-space cues such as optical flow [14, 26], and a forecast of the future [26, 36, 53]. Video-based WAMs forecast whole future frames or latents at uniform fidelity [40, 46], and conditioning on motion cues [26] or restricting prediction to dynamic regions and salient tokens [36, 53] keeps the forecast in image space. These image-space forecasts spend computation on visual detail, while the quantity that decides contact, the target's metric pose at execution time, stays implicit. Target-centric predictors and dynamic VLAs with object-level forecasts sit at the other end [14, 17, 27]: they are fast but blind to the surroundings that can block or redirect the motion. Neither approach allocates precision according to what contact requires from each part of the scene. This raises a central question: what must a policy know about a moving scene, and how precisely, to make contact while keeping inference latency low?

Our key observation is that a dynamic policy needs to know very little precisely. Contact demands millimeter precision in the target's pose at execution time; the surroundings matter only through the clearance they leave and the motion they impart to the target, so they need only be known coarsely. Building on this observation, we introduce SE(3)-WAM, a world–action model that learns a role-aware metric 4D representation, precise where contact demands it (Fig. 1). To supply motion information without encoding past image frames, SE(3)-WAM pairs a single RGB-D frame with a short history of the target's metric SE(3) poses. A world stream learns the representation by forecasting the target's future SE(3) trajectory and the motion of the whole scene as a sparse 3D flow. An action stream conditions on the world stream's learned summary to generate action chunks, with explicit forecasting confined to training [50]. SE(3)-WAM thus combines precise anticipation of target motion with low inference latency, limiting how much the scene changes between observation and action.

We build Dynamic-14, an open-source benchmark of 14 simulated dynamic manipulation tasks in which the scene keeps moving throughout each policy's inference, so every method pays for its own latency. SE(3)-WAM achieves [XX.X]% success on Dynamic-14, [XX.X] points above the strongest baseline, at a p95 latency of [XX] ms, [N]× lower than that of video world–action baselines. It also

leads on the 24 tasks of DOMINO [14] and succeeds in [XX]% of trials on six real-robot tasks, against [XX]% for the strongest baseline. Controlled ablations test the contributions of target-motion history and future supervision, and compare role-aware with uniform-fidelity forecasts to examine the allocation of precision.

Our contributions are threefold:

- We show that a dynamic policy needs millimeter precision only in the target's pose at execution time, and instantiate this principle as role-aware 4D modeling: a precise metric SE(3) trajectory for the target and a coarse sparse 3D flow for the whole scene.
- We develop SE(3)-WAM, a world–action model that combines metric target-pose histories with role-aware predictive supervision for low-latency action generation.
- We introduce Dynamic-14, an open-source dynamic manipulation benchmark that charges each policy its own inference latency, and demonstrate gains over strong baselines on Dynamic-14, 24 DOMINO tasks, and six real-robot tasks.

2. Related Work

Dynamic manipulation. Compensating for the time between sensing and acting is a classical problem, from control with dead time [33] to catching systems that track the target with Kalman-type filters and plan an interception [2–4, 29]; such systems are precise but engineered for each object and task. Learned policies have recently been brought to moving scenes [1]. PUMA reads historical optical flow and is supervised with future object features [14], DynamicWAM conditions a WAM on flow and kinematic descriptors and executes it with real-time chunking [26], and other methods forecast salient tokens, future images, or compact future codes and feed them to the policy at test time [27, 36, 51, 54]; DynaWM refines the chunks of a base VLA for moving objects [11], and DynamicManip synthesizes dynamic demonstrations from a static one [22]. Execution schemes instead reduce or hide latency, by overlapping inference with execution [31, 44], keeping consecutive chunks consistent [8, 24], or conditioning on the execution time [37]. They decide when actions are computed and applied, whereas SE(3)-WAM changes what the policy represents; we run every method under the same real-time executor and, as a control, with its original executor. DOMINO [14] and PhysMani-Bench [51] provide diverse dynamic-manipulation tasks, but their released default evaluation pipelines advance the simulation only after a policy has returned its prediction; Dynamic-14 advances the scene during inference and charges each method its measured latency.

Predictive supervision in world–action models. World–action models (WAMs) generate future video and actions in one network [21, 40, 46, 58], building on video prediction as a policy objective [38, 43]. Because video is costly and dominated by appearance, recent work changes what is forecast, such as dynamic regions, depth, and semantic features [53], flow and geometry alongside RGB [49], object masks or slots [25, 47], or compact latents [57], and asks whether the forecast is needed at test time: Fast-WAM retains most of the benefit of video modeling without test-time imagination [50], and PUMA and others drop their predictors after training [14, 35, 55]. SE(3)-WAM adopts this training-only use and changes the target of the supervision: instead of pixels, features, or latents at one resolution, a metric 4D state whose precision follows each entity's role in contact.

Geometric representations of motion. Metric 3D inputs make policies more precise and sample-efficient [30, 52]. Motion has been represented by 2D point tracks [5, 42, 45], 3D flow [15, 18, 48], and object keypoints or SE(3) trajectories [17, 28, 34], and rigid-body decompositions have long been used to learn scene dynamics [9, 10]. These works describe the scene with one kind of representation, dense flow or tracks everywhere or a pose for each object; DOCIR separates the robot, the target, and obstacles, but as input encodings [13]. SE(3)-WAM assigns precision by role: a coarse sparse 3D flow describes the whole scene, and the target, on which contact depends to within millimeters, is additionally described by its SE(3) history and forecast.

3. SE(3)-WAM

SE(3)-WAM is a world–action model that represents a dynamic scene at role-dependent precision (Fig. 1). It encodes the current RGB-D frame, whose RGB a frozen MAE ViT-B/16 [16] turns into 196 patch tokens and whose depth gives each patch the metric 3D position of its center point in the robot base frame, together with short histories of the target's SE(3) pose and of the robot state. These 212 scene tokens and 16 learnable summary tokens that aggregate them form a world stream, which forecast heads supervise during training with the target's future SE(3) trajectory and a sparse 3D flow of the whole scene. An action stream shares attention with the world stream, reads its summary tokens, and generates action chunks by conditional flow matching [23].

3.1. Problem Formulation

SE(3)-WAM learns a policy from demonstrations. At query t, the policy receives an RGB image, a depth map, and the robot state captured at time τ_t, together with short histories of the robot state and of the pose $T(\tau) = (R(\tau), \mathbf{p}(\tau)) \in \mathrm{SE}(3)$ of a target object in the robot base frame. It outputs a chunk $(\mathbf{a}_{t,0}, \ldots, \mathbf{a}_{t,H-1})$ of commands, one per control period Δ_a, indexed by time from capture: the j-th command is meant for

$$\tau^{\mathrm{a}}_{t,j} = \tau_t + j\Delta_a + d_{\mathrm{task}}. \quad (1)$$

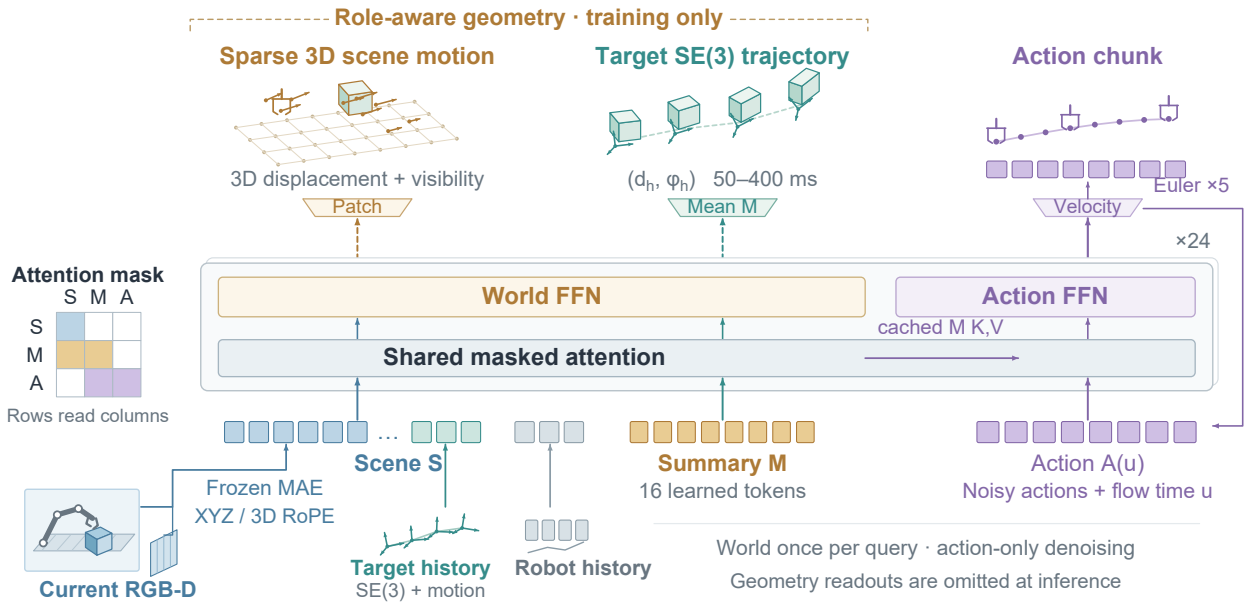


Figure 1. **SE(3)-WAM architecture.** Read from bottom to top. Current RGB-D and target/robot histories feed the world stream. World and action streams share attention and use separate FFNs. Training-only heads (dashed) predict sparse whole-scene 3D displacement and visibility from patch states, and target SE(3) trajectories from the mean summary. Actions read summary tokens through the structured mask. At inference, world keys and values are computed once per query and reused across 5 Euler steps; forecast heads are skipped. Geometry drawings are schematic.

Here d_{task} is a task's built-in command delay. The chunk becomes available only after a latency ℓ_t, so its commands with $j\Delta_a < \ell_t$ have expired, and the later ones execute at their times $\tau^{\text{a}}_{t,j}$. A command must therefore fit the target's pose at $\tau^{\text{a}}_{t,j}$ rather than at τ_t, a pose that the policy must estimate from observations at τ_t, and the lead $\Delta = \tau^{\text{a}}_{t,j} - \tau_t$ of every executed command is at least ℓ_t, so it grows with every millisecond of inference. The bodies of a scene play different roles. The *target* is the body whose pose must be matched within a contact or containment tolerance ε, such as an object to grasp or a receptacle to place into. The *context* consists of the bodies that constrain or drive the target's motion: obstacles, such as an occluder or a wire, must be avoided with a clearance m, and drivers, such as a conveyor, a ramp, or a striker, move the target. The rest of the scene does not affect the task. Sec. 3.2 traces how the error of this estimate depends on the lead and on these roles, and derives the representation of SE(3)-WAM from it.

3.2. Role-Aware 4D Representation

What must be precise. Let a policy act, explicitly or implicitly, on an estimate of the target pose at τ^{a} with position error e and rotation error θ. A contact point at lever arm r is then off by at most $e + 2r\sin(\theta/2)$. Writing the target's motion over the lead as a constant velocity $\mathbf{v}$ plus a residual (acceleration, rolling, impacts), and the lead that the policy assumes as $\hat{\Delta}$, gives

$$e \le e_{\text{loc}} + \hat{\Delta}\, e_v + \|\mathbf{v}\|\, e_\tau + e_{\text{dyn}}, \tag{2}$$

with the errors e_{loc} of the current position, e_v of the velocity, $e_\tau = |\hat{\Delta} - \Delta|$ of the timing, and e_{dyn} of the residual motion (Appendix A.1). The tolerance ε is typically a few millimeters, and the rotation error is amplified by the lever arm, so the target's pose and motion must be known precisely. The context needs far less precision. An obstacle only has to be avoided with the clearance m, which is usually far larger than ε, and a driver matters through the motion it imparts to the target, which then appears in the target's own future; in both cases, a coarse estimate of the context suffices (Fig. 2c). We therefore represent the target by metric SE(3) states, both its observed history and its forecast trajectory, and the whole scene by a sparse 3D flow, all in the robot base frame and in physical time.

What latency allows. Latency enters Eq. (2) twice, through the lead that multiplies every velocity error and through the motion that the residual accumulates. By Eq. (2), information that costs inference time pays off only if it removes more error than the added latency creates. SE(3)-WAM keeps its inputs cheap, with one current image and low-dimensional state histories instead of past frames, and uses its forecasts only to supervise training, so that they add no inference time (Sec. 3.4).

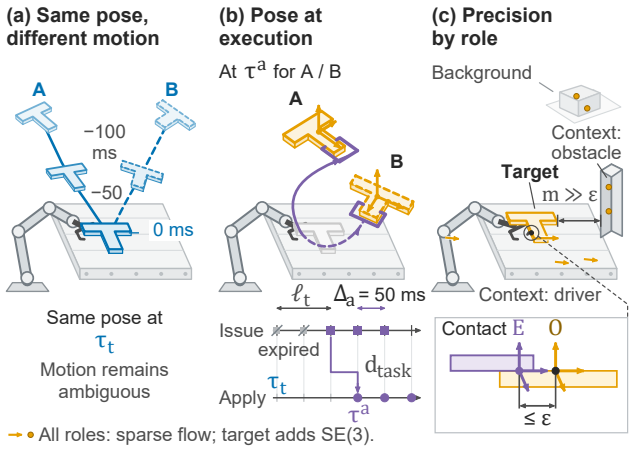


Figure 2. **Acting on a moving target** (schematic). (a) Different histories reach the same captured pose at τ_t. (b) At the same execution time τ^{a}, target translation and rotation differ, requiring different commands (purple). Issue slots are spaced by $\Delta_a = 50$ ms from capture. Gray slots expire during latency ℓ_t; retained commands apply after task delay d_{task}, first at $\tau^{\mathrm{a}} = \tau_t + \ell_t + d_{\mathrm{task}}$. (c) Contact aligns E and O within millimeter-scale ε; an obstacle leaves clearance $m \gg \varepsilon$, a conveyor drives the target, and distant background does not affect the task. Sparse flow covers all roles (arrows: displacement; dots: zero); the target additionally receives precise SE(3) supervision. Poses and delays are illustrative ($\ell_t = 2\Delta_a$, $d_{\mathrm{task}} = \Delta_a$); contact is enlarged.

Target motion: history and forecast. A single pose does not reveal how the target moves (Fig. 2a). The target history covers the last 8 control steps (350 ms at 20 Hz). It consists of the current target pose in the base frame and the seven interval motions between consecutive samples, each with its translation, relative rotation, time interval, and finite-difference velocity and acceleration in physical units, and forms 8 target-state tokens (Appendix A.2). These tokens state directly the velocity and acceleration on which e_v and part of e_{dyn} depend, at the cost of a few low-dimensional tokens. The robot history adds 8 end-effector state tokens. In simulation, the target pose comes from the simulator; on the real robot, a causal tracker estimates it from the RGB-D stream (Appendix D).

The forecast describes where the target will be when the commands execute (Fig. 2b). It covers fixed physical times $\tau_t + \delta_h$, with $\delta_h = h{\cdot}50$ ms and $h = 1, \ldots, 8$, from 50 ms to 400 ms after capture, so that every command time $\tau^{\mathrm{a}}_{t,j}$ with $j \geq 1$ is a forecast horizon. Like all our motion labels, it is cumulative from the current state, and every loss averages over valid labels only. For each horizon, the target's motion is

$$\mathbf{d}_h = \mathbf{p}(\tau_t{+}\delta_h) - \mathbf{p}(\tau_t), \quad \boldsymbol{\phi}_h = \mathrm{Log}\big(R(\tau_t{+}\delta_h)R(\tau_t)^\top\big)^\vee. \tag{3}$$

A linear pose head on the mean of the final summary tokens predicts $(\hat{\mathbf{d}}_h, \hat{\boldsymbol{\phi}}_h)$ with the loss

$$\mathcal{L}_P = \frac{1}{|\mathcal{V}|} \sum_{h \in \mathcal{V}} \Big[\|\hat{\mathbf{d}}_h - \mathbf{d}_h\|^2 + \lambda_R \, d_{\mathrm{SO}(3)}\big(\hat{R}_h, R_h\big)^2 \Big], \tag{4}$$

where $\hat{R}_h = \mathrm{Exp}(\hat{\boldsymbol{\phi}}_h^{\wedge}) R(\tau_t)$ and $\mathcal{V}$ is the set of valid horizons. Because the pose head reads the summary tokens, the only world tokens that the actions attend to (Sec. 3.3), the precise target supervision shapes exactly the representation that the actions use. The past and the future of the target thus share one metric frame and one time axis.

Scene motion: sparse 3D flow. The center points of the 196 patches on the 14×14 grid cover the target, the robot, and the surroundings. For each point i and horizon h, a linear head on its final patch token predicts the base-frame displacement $\hat{F}_{ih} \in \mathbb{R}^3$ and a visibility logit $\hat{o}_{ih}$, and

$$\mathcal{L}_F = \frac{1}{|\mathcal{V}_F|} \sum_{(i,h) \in \mathcal{V}_F} \Big[\tfrac{1}{3} \big\| (\hat{F}_{ih} - F_{ih})/s \big\|^2 + \mathrm{BCE}(\hat{o}_{ih}, o_{ih}) \Big], \tag{5}$$

with a fixed scale s fitted on the training labels, the visibility o_{ih}, and the set $\mathcal{V}_F$ of valid labels. The grid samples the workspace sparsely, about [X] mm between points. The flow thus gives the context and the robot a motion estimate at this spacing and stays near zero on the static background, while the target additionally receives the dedicated SE(3) supervision of Eq. (4). The flow supervises the patch tokens from which the summaries draw, one level below the precise target supervision; this is how SE(3)-WAM allocates precision by role.

Auxiliary supervision. Two further targets regularize the representation; we selected them on development scenes (Appendix E). Future appearance heads on the patch tokens predict, at every horizon, the metric depth, the features of a frozen DINOv3 teacher [32], and the RGB of moving surfaces. Separately, a small head on the summary tokens reconstructs the RGB of the current target region, which ties the summaries to the target without a future label. Their losses form $\mathcal{L}_{\mathrm{aux}}$ (Appendix A.2).

3.3. Action Generation and Joint Learning

Action stream. The action stream holds $H = 8$ tokens, one per command of the chunk; each embeds a noisy command, an end-effector translation and rotation increment plus a gripper command (Appendix A.2). The attention mask $\mathcal{G}$ lets scene tokens attend to the scene, summary tokens to the scene and the summaries, and action tokens to the summaries and the actions, so the actions read the world through the summary tokens and the world stream does not depend on

the actions. In every layer, both streams share the attention projections, through which they exchange information, and keep separate feed-forward networks, which give each its own capacity; we selected this backbone over fully separate parameters on development scenes (Appendix E.4). World tokens use layer normalization, and action tokens use adaptive layer normalization conditioned on the flow time.

Joint learning. For a chunk $\mathbf{a}_1$ normalized with training statistics (Appendix A.2), noise $\mathbf{a}_0 \sim \mathcal{N}(0, I)$, and flow time $u \sim \mathcal{U}[0, 1]$, the action stream v_ψ, with parameters ψ, regresses the flow-matching velocity

$$\mathcal{L}_A = \mathbb{E} \left\| v_\psi(\mathbf{a}_u, u \mid \mathcal{S}_t, \mathcal{G}) - (\mathbf{a}_1 - \mathbf{a}_0) \right\|^2, \qquad (6)$$

with $\mathbf{a}_u = (1-u)\mathbf{a}_0 + u\mathbf{a}_1$, the scene tokens $\mathcal{S}_t$ of query t, and the error averaged over valid commands. All parts are trained jointly with

$$\mathcal{L} = \mathcal{L}_A + \lambda_P \mathcal{L}_P + \lambda_F \mathcal{L}_F + \lambda_{\text{aux}} \mathcal{L}_{\text{aux}}, \qquad (7)$$

with the weights given in Sec. 4.1.

3.4. Efficient Inference and Execution

SE(3)-WAM trains its forecast heads jointly and skips them at test time, as recent WAMs do [21, 50]. Because world tokens attend neither to the actions nor to the flow time, the world stream runs once per query, and the action stream integrates 5 Euler steps on its cached keys and values. The executor counts latency in whole control periods, as RTC does [8]. A query of total latency L_t hands its chunk over at the first control-period boundary at or after it ends, $\ell_t = k_t \Delta_a$ after capture with $k_t = \lceil L_t / \Delta_a \rceil$, and execution continues from command k_t, the one meant for that boundary. There the policy observes again and starts the next query; until then, the executor continues the previous chunk. The policy therefore replans as often as its latency allows, before every command when a query fits in one control period. Because each command is issued at the time its label refers to, the action stream needs no latency input, and the timing error e_τ of Eq. (2) is limited to actuation delays. Every executed command, at most $(H-1)\Delta_a + d_{\text{task}}$ after capture, lies within the horizons of the forecast supervision (Appendix B.3). On the robot, L_t also includes tracking the target in the newest frame.

4. Experiments

We evaluate how SE(3)-WAM performs when the scene keeps moving during inference (Sec. 4.2), what each of its components contributes (Sec. 4.3), and how it transfers to a real robot (Sec. 4.4).

4.1. Setup

Dynamic-14. Dynamic-14 is a simulated benchmark in which observation, scene motion, and action execution share one physical clock, so the scene keeps moving while a policy computes and every method pays for its own latency. It contains fourteen tasks in MuJoCo/robosuite [39, 59] in five families: grasping a moving target (harmonic, conveyor, translating, rotating, and occluded motion), placing into a moving receptacle, passing a wand along a rotating wire, capturing rolling or sliding objects, and grasping after an impact (Fig. 3; Appendix B). Experts with privileged simulator state provide 800 demonstrations per task. We select configurations on development scenes and report all Dynamic-14 results on 1,400 held-out scenes (100 per task, new seeds).

Evaluation. During each query, physics advances by the method's own inference latency, drawn from its latency distribution measured at batch size 1 on one RTX 5090, the GPU of our robot, independently of the hardware that runs the rollouts (Appendix B.3). All methods share one executor that indexes each chunk by time from capture (Sec. 3.4): a query starts at a control-period boundary, and its chunk takes over at the first boundary at or after the query ends (Appendix B.3). Besides this native-latency setting, we add 150 ms of latency to every method and pause physics during inference; Appendix C.2 adds equal-latency and original-executor controls. We report the success rate (SR), averaged over the fourteen tasks with equal weight and over training seeds, and the p95 latency, the measured time from an observation to its first command (time-to-action). Paired 95% intervals come from a bootstrap over held-out scenes.

Baselines. We compare with reactive policies, dynamic VLAs, image- or latent-space WAMs, and a dynamic WAM (Table 1). Each baseline is trained on the same demonstrations with its official code, inputs, and recipe, including a rendered wrist camera and frame or flow histories where the recipe uses them (Appendix B.5). SE(3)-WAM uses one agent view with its depth and the target's SE(3) history, without past frames. To give baselines the same target-motion information, $\pi_{0.5}$ and DynamicWAM also run with our SE(3) history tokens (+SE(3)).

Implementation. The transformer has 24 layers of width 1024 with 16 heads and 523.8M trainable parameters; the MAE ViT-B/16 encoder and the DINOv3 teacher stay frozen. On Dynamic-14, it trains from random initialization for 20 epochs with AdamW, $\lambda_P = \lambda_F = 1$, and $\lambda_R = \lambda_{\text{aux}} = 0.1$ (Appendix A.3). On the real robot, it is fine-tuned per task from a 48-layer base pretrained on DROID [19] (Appendix A.4).

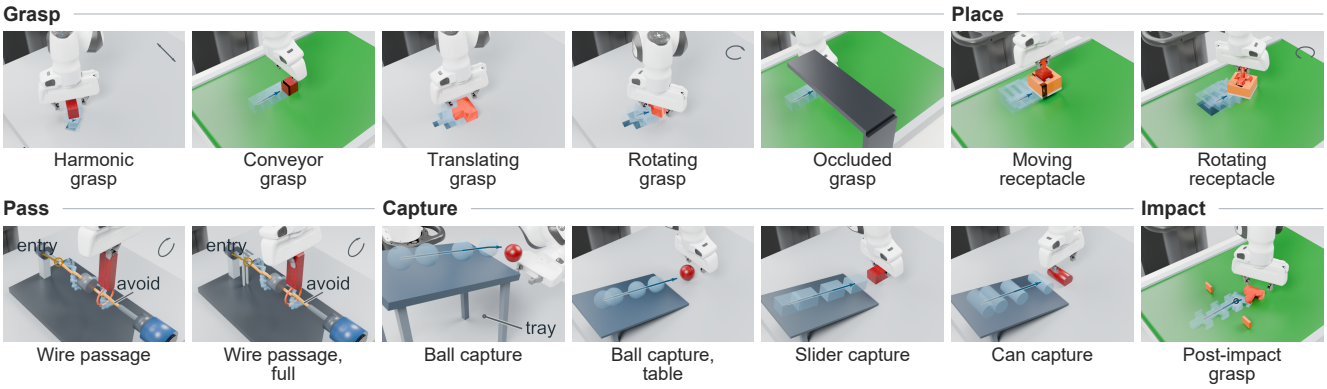


Figure 3. **Dynamic-14.** Fourteen tasks in five families, rendered from recorded simulator states with a common agent view. Four translucent blue poses fade from recent to earlier history; navy traces follow measured past motion. Gray corner arrows indicate rotation or oscillation qualitatively. A small ring marks the target center during recorded impact. Both Pass panels show passage: the wire is a target at entry and an obstacle along the path; Full additionally requires acquisition, exit and release. Occluded motion stays hidden. These native-expert examples illustrate task dynamics, not SE(3)-WAM outcomes; no future prediction is shown.

Table 1. **Main results.** Task-balanced SR on 1,400 held-out Dynamic-14 scenes: native latency, +150 ms, and paused physics. Three training seeds for SE(3)-WAM, one per baseline. p95: time-to-action on RTX 5090. Wrist and history denote extra inputs (Sec. 4.1). Blue/bold and light blue mark the best and second reported values per column; rankings remain provisional while results are pending.

Method	History: Wrist RGB	History: Flow	History: SE(3)	SR (%) ↑ Native	SR (%) ↑ +150 ms	SR (%) ↑ Paused	p95 (ms) ↓
ACT [56]				**42.0**	**24.1**	**40.9**	**10.7**
DP [12]		✓		0.0	0.0	33.1	1701.9
$\pi_{0.5}$ [6]				—	—	—	—
$\pi_{0.5}$+SE(3)	—		✓	—	—	—	—
DynamicVLA [44]		✓		—	—	—	—
PUMA [14]		✓		—	—	—	—
OpenWAM-α [40]				—	—	—	—
DynamicWAM [26]		✓		—	—	—	—
DynamicWAM+SE(3)	—	✓	✓	—	—	—	—
SE(3)-WAM			✓	—	—	—	—

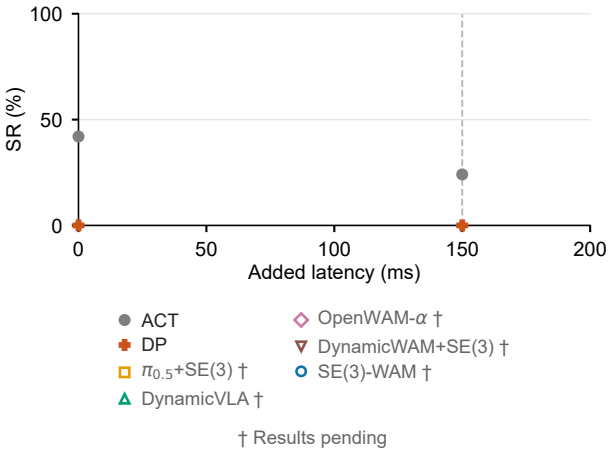


Figure 4. **Success under added latency.** Comparison on 1,400 held-out Dynamic-14 scenes. The legend retains all planned series, plus DP; † marks methods awaiting results. ACT and DP have measurements at 0/150 ms. Only adjacent measured points are connected. Dashed line: +150 ms; full sweep table: Table S7.

4.2. Simulation Results

Comparison. With its native latency, SE(3)-WAM reaches [XX.X]% SR on held-out Dynamic-14 scenes (Table 1), [X.X] points above the strongest baseline, [name], and [X.X] points above the strongest baseline that receives the same SE(3) history, [name]. Its gains are largest on [task types] (Appendix C.1). Its p95 latency of [XX] ms is [X]× lower than that of the image-space WAMs and close to that of [the reference reactive policy, to be fixed].

Latency and protocols. With 150 ms of added latency, SE(3)-WAM keeps [XX.X]% SR, whereas the reactive policies drop to [XX.X]% and the image-space WAMs to [XX.X]% on average, as Eq. (2) predicts: added latency lengthens the lead that multiplies every velocity error, so methods that do not anticipate the target's motion lose most. Fig. 4 shows the available measurements in the 0–200 ms sweep; pending points are listed in Appendix C.2. With physics paused during inference, which ignores the cost of computation, SE(3)-WAM still leads [name] by [X.X] points, against [X.X] points with native latency. When every method is charged the p95 latency of SE(3)-WAM, its margin

Table 2. **DOMINO results** on 24 single-arm clean-L1 tasks, each method trained on DOMINO's demonstrations (one training seed per method, 50 episodes per task and method). SR (%) is grouped by interaction type and averaged over tasks; MS is the manipulation score. Best per column will be bold on blue once measured. Protocol: Appendix C.3.

Method	Intercept	Track	All ↑	MS ↑
$\pi_{0.5}$	—	—	—	—
DynamicWAM	—	—	—	—
PUMA	—	—	—	—
SE(3)-WAM	—	—	—	—

Table 3. **Real-robot results.** Success rate (%) on the six tasks of Sec. 4.4, 20 trials per task and method; All pools the trials. p95 lat.: p95 latency, the end-to-end time-to-action (ms) on one RTX 5090, including tracking for SE(3)-WAM. DP: Diffusion Policy. Once measured, best per column will be bold on blue, including ties. Successes, intervals, and failures in Appendix D.

Method	T1	T2	T3	T4	T5	T6	All ↑	p95 (ms) ↓
$\pi_{0.5}$	—	—	—	—	—	—	—	—
DynamicWAM	—	—	—	—	—	—	—	—
DP	—	—	—	—	—	—	—	—
SE(3)-WAM	—	—	—	—	—	—	—	—

is [X.X] points, and running [the baselines that rely on their own asynchronous execution] with their original executors changes their SR by at most [X.X] points (Appendix C.2).

DOMINO. We also train SE(3)-WAM, $\pi_{0.5}$, DynamicWAM, and PUMA on 24 single-arm Franka tasks of DOMINO [14] with its demonstrations and evaluate them with its native evaluator. SE(3)-WAM reaches [XX.X]% SR against [XX.X]% for [the strongest baseline] (Table 2; Appendix C.3). As on Dynamic-14, it reads the simulator pose of the target.

4.3. Ablations and Analysis

Five controls each change one component of the final model and keep the data, recipe, and training budget, with three training seeds each (Fig. 5). They test the argument of Sec. 3.2: the target history supplies the velocity on which e_v in Eq. (2) depends, future supervision shapes the representation without adding inference time, and the two geometric components and a uniform-fidelity forecast test the allocation of precision by role. We evaluate them with native latency and with 150 ms of added latency (per-seed results and a sensitivity analysis in Appendix C.4); since the forecast heads are skipped at deployment, all controls share the inference cost of SE(3)-WAM.

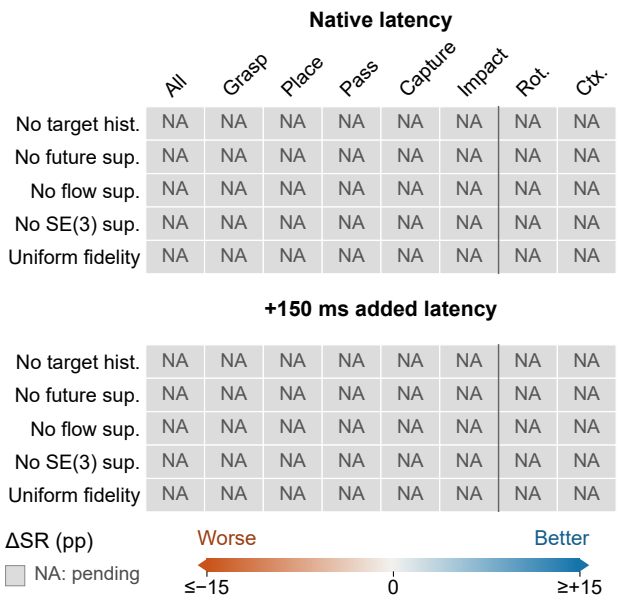


Figure 5. **Ablation effects by task (design preview; data pending).** ΔSR = SR(control) − SR(full), in percentage points; colors saturate at ±15. Rows: the five controls of Sec. 4.3; the full model reaches [XX.X]% SR with native latency and [XX.X]% with +150 ms. Rot./Ctx.: predeclared, overlapping rotation-sensitive/context-active groups (Table S3). Planned estimates average three seeds per scene and weight tasks equally, retaining `t3_phase`. Bold will mark unadjusted paired 95% scene-bootstrap intervals excluding zero, conditional on trained weights (Appendix B.4). Grey NA: no eligible paired results or estimates.

Target history. The no-history control removes the past target poses with their velocity and acceleration features and keeps the current image, the current target pose, and the robot history; unlike the other controls, it changes the input. Without the history, SR drops by [X.X] points with native latency and by [X.X] points with 150 ms of added latency, as Eq. (2) predicts: a longer lead amplifies the velocity error that the history removes. Because the history relies on pose estimates, we also perturb them; under synthetic correlated pose noise, SE(3)-WAM keeps [XX.X]% SR (Appendix C.5).

Future supervision. The no-future control removes all forecast losses, the SE(3), scene-flow, and future appearance targets, and keeps the current-target RGB reconstruction. Its inference is identical to that of SE(3)-WAM, yet SR drops by [X.X] points with native latency and by [X.X] points with 150 ms of added latency, most on [tasks with non-uniform target motion]. Offline, the learned SE(3) forecast is also more accurate than constant-velocity and constant-acceleration extrapolation of the same history (Table S9).

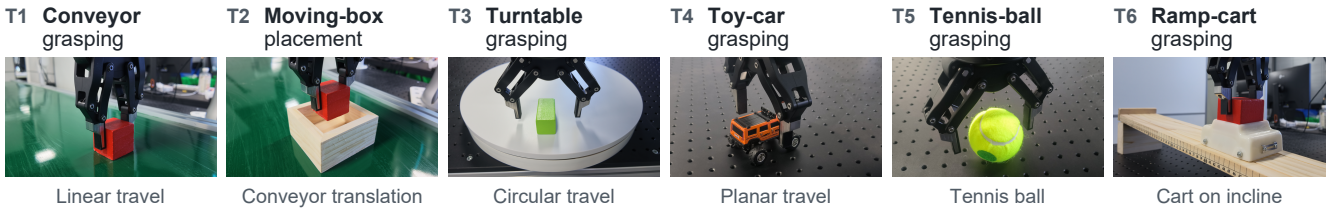


Figure 6. **Real-robot task setups.** Photographs of conveyor grasping (T1), placement into a moving box (T2), turntable grasping (T3), moving toy-car grasping (T4), tennis-ball grasping (T5), and grasping a block carried by a ramp cart (T6).

Allocation of precision. Three controls keep all auxiliary targets and change the geometric forecast. Removing the scene-flow supervision lowers SR by [X.X] points; because the flow also covers the target and the robot, and the appearance targets remain, this measures the contribution of coarse scene-motion supervision rather than of all context information. Removing the dedicated SE(3) supervision leaves the target with only the coarse flow that every part of the scene receives and lowers SR by [X.X] points. If precision follows the roles, the SE(3) supervision should matter most on tasks with a tight pose tolerance or a rotation-sensitive contact, and the flow supervision on tasks whose context constrains or drives the target. [Whether Fig. 5 shows this pattern; if it does not, state that the controls establish the contribution of each component but not the allocation by role.] The uniform-fidelity control replaces the SE(3) and scene-flow targets by [future latents of a frozen video VAE] of the whole frame at the same horizons, so that every part of the scene is forecast at one fidelity; at the same inference cost, it trails SE(3)-WAM by [X.X] points. [One-sentence summary of which components matter, and where.]

4.4. Real-Robot Experiments

Setup. We evaluate six tasks on a Franka arm ([model]) with [gripper model] and a fixed RealSense D435 RGB-D camera (Fig. 6; Appendix D): conveyor grasping (T1), placement into a moving receptacle (T2), turntable grasping (T3), moving toy-car grasping (T4), tennis-ball grasping (T5), and block grasping from a ramp cart (T6). Baselines are $\pi_{0.5}$ (VLA), DynamicWAM (dynamic WAM), and Diffusion Policy [12] (reactive). All methods train or fine-tune on the same 100 teleoperated demonstrations per task and run 20 trials per task with shared trial keys (Appendix D.3). Baselines receive RGB [and a wrist camera if their recipe uses one]; SE(3)-WAM also receives FoundationPose [41] target estimates, updated at every control-period boundary before its query (Appendix D.2). End-to-end latencies include tracking (Table 3).

Results. SE(3)-WAM succeeds in [XX.X]% of all trials, against [XX.X]% for [the strongest baseline], a paired difference of [X.X] points (95% interval [[X.X, X.X]]; Table 3). Its p95 latency, including tracking, is [XX] ms, [within one control period], whereas [name] needs [XX] ms. The margin is largest on [tasks], and the dominant failure of SE(3)-WAM is [failure mode] (Appendix D.4).

5. Conclusion

SE(3)-WAM spends precision where contact depends on it: a metric SE(3) history and forecast of the target and a sparse 3D flow of the whole scene, with the forecasts used only as training supervision. It reaches [XX.X]% success on Dynamic-14, [XX.X]% on DOMINO, and [XX]% on six real-robot tasks at a p95 latency of [XX] ms, improving on reactive, dynamic, and world–action policies. [The ablations show that the target history matters most under added latency and that future supervision improves control without changing the inference computation.] [State what the dedicated SE(3), scene-flow, and uniform-fidelity controls support: a better forecast design, or, if the task groups also agree, precision allocated by role.] Accounting for inference latency [state how success rates and relative performance differ from paused evaluation].

Limitations. SE(3)-WAM requires a specified rigid target and reliable estimates of its pose, and its forecast does not model how the robot's own contact changes the target's motion. Its target-trajectory head predicts a single future per horizon and does not represent uncertainty or multimodal motion, which an uncertainty-aware forecast could address.

LLM Usage Statement

We used OpenAI Codex to assist with literature review, code implementation and debugging, experiment execution, and language editing of the manuscript. Codex also assisted in developing plotting code and refining figure and manuscript layouts. The authors reviewed AI-assisted content, checked references against original sources, and validated the implementation through testing and experiments. The authors take full responsibility for the final manuscript, experimental results, and conclusions.

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